

# BS-PH101 — Physics-I

Semester I/II • 3L:1T:0P • 4 Credits

## Module 1 Comprehensive Exam Notes: Mechanics

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. Constraints and Friction

### 1.1 Constraints and Degrees of Freedom

#### Definition: Constraint and Degrees of Freedom

A **constraint** is any physical restriction on the position or motion of a mechanical system. The number of independent coordinates required to completely specify the configuration of a system is called its **Degrees of Freedom (DOF)**.

If a system consists of  $N$  particles in 3D space subject to  $k$  independent constraint equations, the number of degrees of freedom is given by:

$$f = 3N - k \quad (1)$$

#### Classification of Constraints:

| Constraint Type      | Condition / Mathematical Form                                                                                | Physical Example                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Holonomic</b>     | Expressible as $f(r_1, r_2, \dots, r_N, t) = 0$ (integrable algebraic relations).                            | Rigid body ( $ \vec{r}_i - \vec{r}_j  = c$ ), simple pendulum ( $r = L$ ).              |
| <b>Non-Holonomic</b> | Expressible as inequalities or non-integrable differential relations ( $f(\vec{r}, \dot{\vec{r}}, t) = 0$ ). | Rolling without slipping on a rough plane, gas molecules in a container ( $r \leq R$ ). |
| <b>Scleronomic</b>   | Constraint equations do not explicitly depend on time $t$ ( $\partial f / \partial t = 0$ ).                 | Bead on a fixed wire ring, rigid dumbbell.                                              |
| <b>Rheonomic</b>     | Constraint equations explicitly depend on time $t$ ( $\partial f / \partial t \neq 0$ ).                     | Bead on a wire ring rotating with constant angular velocity $\omega$ .                  |

### 1.2 Friction and Laws of Friction

Friction is the resistive contact force that opposes the relative motion or tendency of relative motion between two surfaces in contact.

#### Key Laws and Relations of Friction

- **Static Friction ( $f_s$ ):** Self-adjusting force. Maximum value is limiting friction:

$$f_{s,\max} = \mu_s N \quad (2)$$

- **Kinetic Friction ( $f_k$ ):** Constant resistive force during motion:

$$f_k = \mu_k N \quad (\text{where } \mu_k < \mu_s) \quad (3)$$

- **Angle of Friction ( $\lambda$ ):** Angle between total contact reaction  $\vec{R}$  and normal reaction  $\vec{N}$ :

$$\tan \lambda = \frac{f_{s,\max}}{N} = \mu_s \implies \lambda = \tan^{-1}(\mu_s) \quad (4)$$

- **Angle of Repose ( $\theta_r$ ):** Maximum angle of inclination of a rough plane at which a body remains at rest without sliding:

$$mg \sin \theta_r = \mu_s mg \cos \theta_r \implies \tan \theta_r = \mu_s \implies \theta_r = \lambda \quad (5)$$

#### Solved Problem 1: Block on an Inclined Plane with Friction

**Problem:** A block of mass  $m = 5$  kg rests on an inclined plane of angle  $\theta = 30^\circ$ . The coefficient of static friction is  $\mu_s = 0.4$ . Determine whether the block slides down, and find the frictional force.

**Solution:**

1. Component of gravity along the incline:  $F_{\text{down}} = mg \sin 30^\circ = 5 \times 9.8 \times 0.5 = 24.5$  N.
2. Normal reaction:  $N = mg \cos 30^\circ = 5 \times 9.8 \times 0.866 = 42.435$  N.
3. Maximum limiting static friction:  $f_{s,\text{max}} = \mu_s N = 0.4 \times 42.435 = 16.97$  N.
4. Since  $F_{\text{down}}(24.5 \text{ N}) > f_{s,\text{max}}(16.97 \text{ N})$ , the block **will slide down**. The friction force acting up the plane during motion is kinetic friction  $f_k = \mu_k N$ .

#### Solved Problem 2: Degrees of Freedom of Connected Bodies

**Problem:** Find the degrees of freedom for: (a) A double pendulum moving in a plane, (b) Two particles connected by a rigid rod free to move in 3D space.

**Solution:**

1. **Double Pendulum in 2D:**  $N = 2$  particles, constrained to plane ( $z_1 = 0, z_2 = 0$ ), fixed length  $r_1 = L_1$  and  $|\vec{r}_2 - \vec{r}_1| = L_2$ . Total independent angles  $\theta_1, \theta_2$ . Hence DOF = 2.
2. **Rigid Dumbbell in 3D:**  $N = 2$  particles in 3D (6 coordinates) subject to  $k = 1$  constraint ( $|\vec{r}_1 - \vec{r}_2| = d$ ). Hence DOF =  $3(2) - 1 = 5$  (3 translational, 2 rotational).

## 2. Vector Calculus and Potential Energy Function

### 2.1 Gradient, Divergence, and Curl

#### Definition: Gradient, Divergence, and Curl

- **Gradient ( $\nabla V$ ):** Vector representing the rate and direction of maximum spatial change of a scalar field  $V(x, y, z)$ .
- **Divergence ( $\nabla \cdot \vec{A}$ ):** Scalar representing the net outward flux of a vector field per unit volume.
- **Curl ( $\nabla \times \vec{A}$ ):** Vector representing the circulation or rotational density of a vector field.

## Mathematical Definitions in Cartesian Coordinates

$$\text{grad } V = \nabla V = \frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \quad (6)$$

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \quad (7)$$

$$\text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \quad (8)$$

2.2 Potential Energy Function  $F = -\text{grad } V$ 

For a conservative force field  $\vec{F}$ , the work done along a closed path is zero ( $\oint \vec{F} \cdot d\vec{r} = 0$ ). This implies  $\vec{F}$  can be expressed as the negative gradient of a scalar potential energy function  $V(\vec{r})$ :

$$\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}\right) \quad (9)$$

**Physical Meaning of Negative Sign:** The force acts in the direction of *decreasing* potential energy (towards stable equilibrium).

**Equipotential Surfaces:** A surface on which the potential energy  $V(\vec{r}) = \text{constant}$ .

- Work done in moving a mass along an equipotential surface is zero:  $dW = \vec{F} \cdot d\vec{r} = -dV = 0$ .
- The force vector  $\vec{F}$  is everywhere perpendicular to the equipotential surface passing through that point.

## Solved Problem 3: Deriving Force from Potential Energy

**Problem:** Given potential energy  $V(x, y, z) = 3x^2y - y^3z^2$ , calculate force  $\vec{F}$  at point  $(1, -1, 2)$ .

**Solution:**

1. Calculate partial derivatives:  $\frac{\partial V}{\partial x} = 6xy$ ,  $\frac{\partial V}{\partial y} = 3x^2 - 3y^2z^2$ ,  $\frac{\partial V}{\partial z} = -2y^3z$ .
2. Evaluate at  $(1, -1, 2)$ :  $\frac{\partial V}{\partial x} = 6(1)(-1) = -6$ ,  $\frac{\partial V}{\partial y} = 3(1)^2 - 3(-1)^2(2)^2 = 3 - 12 = -9$ ,  $\frac{\partial V}{\partial z} = -2(-1)^3(2) = 4$ .
3. Substitute into  $\vec{F} = -\nabla V$ :

$$\vec{F} = -(-6\hat{i} - 9\hat{j} + 4\hat{k}) = 6\hat{i} + 9\hat{j} - 4\hat{k} \text{ N} \quad (10)$$

## 3. Conservative Forces and Conservation Laws

### 3.1 Conservative vs. Non-Conservative Forces

#### Comparative Breakdown:

| Property          | Conservative Force                                      | Non-Conservative Force                         |
|-------------------|---------------------------------------------------------|------------------------------------------------|
| Work Dependence   | Work done depends only on initial and final positions.  | Work done depends on the actual path taken.    |
| Closed Path Work  | $\oint \vec{F} \cdot d\vec{r} = 0$ .                    | $\oint \vec{F} \cdot d\vec{r} \neq 0$ .        |
| Curl Condition    | $\nabla \times \vec{F} = 0$ (Irrotational).             | $\nabla \times \vec{F} \neq 0$ .               |
| Potential Energy  | Expressible as $\vec{F} = -\nabla V$ .                  | Potential energy function cannot be defined.   |
| Mechanical Energy | Total mechanical energy ( $E = K + V$ ) is conserved.   | Mechanical energy is dissipated as heat/sound. |
| Examples          | Gravitational force, Electrostatic force, Spring force. | Friction, Viscosity, Air resistance.           |

### 3.2 Fundamental Conservation Laws

#### Conservation Theorems

1. **Conservation of Linear Momentum:** If net external force  $\vec{F}_{\text{ext}} = 0$ , then:

$$\frac{d\vec{P}}{dt} = 0 \implies \vec{P} = \sum m_i \vec{v}_i = \text{constant} \quad (11)$$

2. **Conservation of Angular Momentum:** If net external torque  $\vec{\tau}_{\text{ext}} = \vec{r} \times \vec{F}_{\text{ext}} = 0$ , then:

$$\frac{d\vec{L}}{dt} = 0 \implies \vec{L} = \vec{r} \times \vec{P} = I\vec{\omega} = \text{constant} \quad (12)$$

3. **Conservation of Energy:** In an isolated system with conservative forces:

$$E_{\text{total}} = K + V = \frac{1}{2}mv^2 + V(\vec{r}) = \text{constant} \quad (13)$$

## 4. Non-Inertial Frames of Reference

### 4.1 Inertial vs. Non-Inertial Frames

#### Definition: Frames of Reference

- **Inertial Frame:** A frame of reference moving with zero acceleration (at rest or constant velocity) in which Newton's laws of motion hold strictly.
- **Non-Inertial Frame:** An accelerating or rotating frame of reference in which Newton's laws do not hold directly unless fictitious / pseudo forces are introduced.

## 4.2 Rotating Frames: Centrifugal and Coriolis Forces

Consider a frame  $S'$  rotating with constant angular velocity  $\vec{\omega}$  relative to a stationary inertial frame  $S$ .

### Transformation Equation for Acceleration

The transformation relation between acceleration in fixed frame  $\vec{a}_S$  and rotating frame  $\vec{a}_{S'}$  is:

$$\vec{a}_S = \vec{a}_{S'} + 2(\vec{\omega} \times \vec{v}') + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (14)$$

Multiplying by mass  $m$ , the effective force in the rotating frame is:

$$\vec{F}' = m\vec{a}_{S'} = \vec{F}_{\text{real}} - \underbrace{2m(\vec{\omega} \times \vec{v}')}_{\text{Coriolis Force } \vec{F}_{\text{cor}}} - \underbrace{m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centrifugal Force } \vec{F}_{\text{cent}}} \quad (15)$$

### Properties of Rotating Fictitious Forces:

| Force                    | Vector Expression                                                                                           | Key Characteristics                                                                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Centrifugal Force</b> | $\vec{F}_{\text{cent}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}) = m\omega^2 r_{\perp} \hat{r}$ | Directed radially outward from axis of rotation. Present even if particle is stationary in rotating frame ( $\vec{v}' = 0$ ).                                           |
| <b>Coriolis Force</b>    | $\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$                                                  | Perpendicular to both velocity $\vec{v}'$ and angular velocity $\vec{\omega}$ . Acts only when particle is in motion ( $\vec{v}' \neq 0$ ). Does zero work ( $W = 0$ ). |

### Physical Effects of Coriolis Force on Earth:

- **Deflection of Moving Objects:** Moving bodies in Northern Hemisphere are deflected to their *right*, while in Southern Hemisphere to their *left* (causes cyclone rotation direction).
- **Eastward Deflection of Freely Falling Body:** A body dropped from height  $h$  at latitude  $\lambda$  experiences an eastward deflection  $\Delta x = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}}$ .
- **Foucault Pendulum:** Demonstrates Earth's rotation directly via precession of its oscillation plane at angular rate  $\Omega = \omega \sin \lambda$ .

### Solved Problem 4: Eastward Deflection Calculation

**Problem:** Calculate the eastward deflection due to Coriolis force for a body dropped from a height  $h = 100$  m at Kolkata (latitude  $\lambda = 22.5^\circ\text{N}$ ). (Take  $\omega = 7.29 \times 10^{-5}$  rad/s,  $g = 9.8$  m/s<sup>2</sup>).

**Solution:**

1. Formula for eastward deflection:

$$d = \frac{1}{3}\omega g \cos \lambda \sqrt{\frac{8h^3}{g}} \quad (16)$$

2. Calculate time of fall:  $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{200}{9.8}} = 4.518$  s.

3. Substitute values:  $d = \frac{1}{3}(7.29 \times 10^{-5})(9.8) \cos(22.5^\circ)(4.518)^3 \cos 22.5^\circ = 0.9239$ ,  $(4.518)^3 = 92.23$ .  $d = 0.333 \times 7.29 \times 10^{-5} \times 9.8 \times 0.9239 \times 92.23 = \mathbf{0.0195 \text{ m} = 1.95 \text{ cm}}$ .

## 5. Harmonic Oscillations

### 5.1 Damped Harmonic Motion

When an oscillator experiences a viscous damping force proportional to velocity ( $F_{\text{damping}} = -b\dot{x}$ ), the equation of motion is:

$$m\ddot{x} + b\dot{x} + kx = 0 \implies \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0 \quad (17)$$

where  $\gamma = \frac{b}{2m}$  is the damping coefficient and  $\omega_0 = \sqrt{\frac{k}{m}}$  is the natural frequency.

Assuming solution  $x(t) = Ce^{rt}$ , the roots of characteristic equation are  $r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$ .

#### Three Cases of Damped Motion

1. **Overdamped** ( $\gamma > \omega_0$ ): Non-oscillatory exponential return to equilibrium.

$$x(t) = e^{-\gamma t} \left( Ae^{\sqrt{\gamma^2 - \omega_0^2} t} + Be^{-\sqrt{\gamma^2 - \omega_0^2} t} \right) \quad (18)$$

2. **Critically Damped** ( $\gamma = \omega_0$ ): Fastest non-oscillatory return to equilibrium without overshoot. Used in shock absorbers and deadbeat galvanometers.

$$x(t) = (A + Bt)e^{-\gamma t} \quad (19)$$

3. **Underdamped** ( $\gamma < \omega_0$ ): Oscillatory motion with exponentially decaying amplitude.

$$x(t) = A_0 e^{-\gamma t} \cos(\omega_d t + \phi), \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \gamma^2} \quad (20)$$

#### Damping Parameters: Logarithmic Decrement and Quality Factor

- **Relaxation Time** ( $\tau$ ): Time in which energy drops to  $1/e$  of initial value:  $\tau = \frac{1}{2\gamma} = \frac{m}{b}$ .
- **Logarithmic Decrement** ( $\delta$ ): Natural logarithm of ratio of successive amplitudes one period  $T_d$  apart:

$$\delta = \ln \left( \frac{A_n}{A_{n+1}} \right) = \gamma T_d = \frac{b T_d}{2m} \quad (21)$$

- **Quality Factor** ( $Q$ ): Dimensionless measure of sharpness of oscillation energy retention:

$$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma} = \frac{\omega_0 \tau}{1} = \frac{\pi}{\delta} \quad (22)$$

### 5.2 Forced Oscillations and Resonance

Under a periodic external driving force  $F(t) = F_0 \cos(\omega t)$ , the differential equation is:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad \left( \text{where } f_0 = \frac{F_0}{m} \right) \quad (23)$$

In steady state, the response oscillates at driving frequency  $\omega$ :

$$x(t) = x_0 \cos(\omega t - \phi) \quad (24)$$

where steady-state amplitude  $x_0$  and phase lag  $\phi$  are given by:

$$x_0(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \quad \tan \phi = \frac{2\gamma\omega}{\omega_0^2 - \omega^2} \quad (25)$$

#### Resonance Conditions

- **Amplitude Resonance Frequency ( $\omega_r$ ):** Frequency at which displacement amplitude  $x_0$  is maximum:

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}, \quad x_{0,\max} = \frac{f_0}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} \approx \frac{f_0}{2\gamma\omega_0} \quad (\text{for small } \gamma) \quad (26)$$

- **Velocity Resonance Frequency ( $\omega_v$ ):** Frequency at which velocity amplitude  $v_0 = \omega x_0$  is maximum:

$$\omega_v = \omega_0 \quad (\text{Natural undamped frequency}) \quad (27)$$

- **Phase at Resonance:** At velocity resonance ( $\omega = \omega_0$ ),  $\tan \phi = \infty \Rightarrow \phi = 90^\circ$  (force and velocity are in phase).

## 6. Rigid Body Dynamics and Moment of Inertia

### 6.1 Rigid Body Motion and Angular Velocity Vector

A rigid body is a system of particles in which inter-particle distances remain constant under external forces ( $|\vec{r}_i - \vec{r}_j| = \text{constant}$ ).

General motion of a rigid body is a combination of translation of center of mass (CM) and rotation about CM. The angular velocity vector  $\vec{\omega}$  is a vector directed along the instantaneous axis of rotation according to the right-hand rule.

Linear velocity of any particle  $i$  at position  $\vec{r}_i$  relative to an axis is:

$$\vec{v}_i = \vec{\omega} \times \vec{r}_i \quad (28)$$

### 6.2 Moment of Inertia and Parallel/Perpendicular Axes Theorems

#### Definition: Moment of Inertia

**Moment of Inertia ( $I$ )** measures the rotational inertia of a body about a given axis. For a continuous mass distribution:

$$I = \sum m_i r_i^2 = \int r^2 dm = mk^2 \quad (29)$$

where  $k$  is the **radius of gyration**.

### Fundamental Theorems of Moment of Inertia

1. **Parallel Axis Theorem:** MI about any axis is equal to MI about a parallel axis through CM plus  $Mh^2$ :

$$I = I_{\text{cm}} + Mh^2 \quad (h = \text{perpendicular distance between parallel axes}) \quad (30)$$

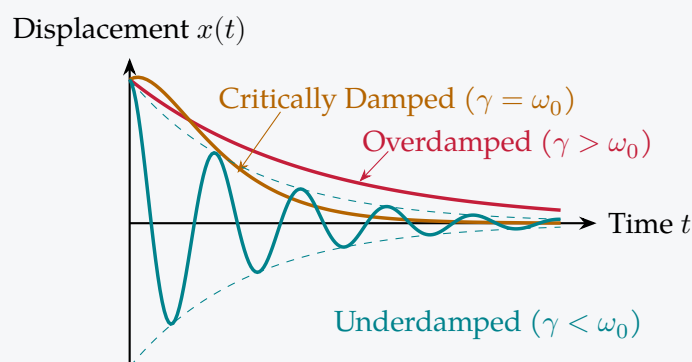
2. **Perpendicular Axis Theorem:** For a planar lamina in  $xy$ -plane, MI about  $z$ -axis perpendicular to lamina is:

$$I_z = I_x + I_y \quad (31)$$

### Standard Moments of Inertia Summary:

| Body                           | Axis of Rotation                     | Moment of Inertia ( $I$ ) |
|--------------------------------|--------------------------------------|---------------------------|
| Thin Uniform Rod (Length $L$ ) | Perpendicular through center.        | $I = \frac{1}{12}ML^2$    |
| Thin Uniform Rod (Length $L$ ) | Perpendicular through one end.       | $I = \frac{1}{3}ML^2$     |
| Circular Ring (Radius $R$ )    | Central axis perpendicular to plane. | $I = MR^2$                |
| Circular Disc (Radius $R$ )    | Central axis perpendicular to plane. | $I = \frac{1}{2}MR^2$     |
| Solid Sphere (Radius $R$ )     | Any diameter axis.                   | $I = \frac{2}{5}MR^2$     |
| Spherical Shell (Radius $R$ )  | Any diameter axis.                   | $I = \frac{2}{3}MR^2$     |
| Solid Cylinder (Radius $R$ )   | Central longitudinal axis.           | $I = \frac{1}{2}MR^2$     |

### Damped Harmonic Oscillator Response Regimes



## Quick Revision — Module 1

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### Key Formulas and Metrics at a Glance

|                                   |                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Degrees of Freedom</b>         | $f = 3N - k$ ( $N$ particles, $k$ constraints)                                                                                                           |
| <b>Limiting Static Friction</b>   | $f_{s,\max} = \mu_s N$ and $\tan \lambda = \mu_s$                                                                                                        |
| <b>Potential Energy Relation</b>  | $\vec{F} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$ |
| <b>Conservative Force Rule</b>    | $\nabla \times \vec{F} = 0 \iff \oint \vec{F} \cdot d\vec{r} = 0$                                                                                        |
| <b>Coriolis Acceleration</b>      | $\vec{a}_{\text{cor}} = 2(\vec{\omega} \times \vec{v}')$ (Deflection to right in Northern Hemisphere)                                                    |
| <b>Underdamped Frequency</b>      | $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ ( $\gamma = b/2m$ )                                                                                            |
| <b>Quality Factor</b>             | $Q = \frac{\omega_0}{2\gamma} = \frac{\pi}{\delta} = \omega_0 \tau$                                                                                      |
| <b>Resonance Amplitude Freq.</b>  | $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$                                                                                                               |
| <b>Parallel Axis Theorem</b>      | $I = I_{\text{cm}} + Mh^2$                                                                                                                               |
| <b>Perpendicular Axis Theorem</b> | $I_z = I_x + I_y$ (Planar lamina in $xy$ -plane)                                                                                                         |

### One-Line Definitions

- **Holonomic Constraint:** A constraint expressible as an integrable algebraic equation of position coordinates and time  $f(\vec{r}, t) = 0$ .
- **Conservative Force:** A force field in which work done along any closed path is zero ( $\oint \vec{F} \cdot d\vec{r} = 0$ ).
- **Coriolis Force:** A velocity-dependent fictitious force acting in a rotating reference frame ( $\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$ ).
- **Logarithmic Decrement:** The natural log of ratio of two consecutive amplitudes one period apart ( $\delta = \gamma T_d$ ).
- **Radius of Gyration:** The perpendicular distance from axis of rotation where whole mass can be assumed concentrated ( $I = Mk^2$ ).

### Frequently Asked / Viva Questions

#### Q1. State the condition for a force field to be conservative.

A force field  $\vec{F}$  is conservative if  $\nabla \times \vec{F} = 0$  (irrotational) or equivalently work done around any closed loop  $\oint \vec{F} \cdot d\vec{r} = 0$ .

#### Q2. Why is Coriolis force called a fictitious force?

Because it arises due to the acceleration of the observer's rotating frame of

## Frequently Asked / Viva Questions (contd.)

reference, not due to physical interactions between bodies.

**Q3. What is the physical significance of negative gradient of potential energy?**

$\vec{F} = -\nabla V$  indicates force always points in direction of steepest decrease of potential energy towards stable equilibrium.

**Q4. Distinguish between holonomic and non-holonomic constraints with examples.**

Holonomic constraints are integrable algebraic position relations (e.g. rigid body length); non-holonomic involve non-integrable velocity relations or inequalities (e.g. sphere rolling without slipping).

**Q5. What is critical damping and state its practical application?**

Critical damping occurs when  $\gamma = \omega_0$ . The system returns to equilibrium in shortest time without oscillating. Used in car shock absorbers and deadbeat galvanometers.

**Q6. How does amplitude resonance frequency differ from velocity resonance frequency?**

Amplitude resonance occurs at  $\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}$  while velocity resonance occurs exactly at natural frequency  $\omega_v = \omega_0$ .

**Q7. What is Quality Factor ( $Q$ ) of an oscillator?**

$Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per cycle}} = \frac{\omega_0}{2\gamma}$ . Higher  $Q$  implies lower damping and sharper resonance.

**Q8. State Parallel and Perpendicular axes theorems of moment of inertia.**

Parallel axis:  $I = I_{\text{cm}} + Mh^2$ . Perpendicular axis:  $I_z = I_x + I_y$  for planar laminas.

**Q9. Why are track rails laid higher on outer curves?**

To provide necessary centripetal force via component of normal reaction and eliminate reliance on lateral friction.

**Q10. Calculate the moment of inertia of a solid sphere of mass  $M$  and radius  $R$  about a tangent.**

By parallel axis theorem,  $I_{\text{tangent}} = I_{\text{cm}} + MR^2 = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$ .

**Q11. What is the total energy of a simple harmonic oscillator of mass  $m$  and amplitude  $A$ ?**

$E = \frac{1}{2}m\omega_0^2 A^2$ . Total mechanical energy is constant and continuously converts between kinetic and potential forms.

**Q12. Show that Coriolis force does no work on a moving particle.**

$\vec{F}_{\text{cor}} = -2m(\vec{\omega} \times \vec{v}')$  is always perpendicular to velocity  $\vec{v}'$ . Power  $P = \vec{F}_{\text{cor}} \cdot \vec{v}' = 0 \Rightarrow W = 0$ .

## Summary Comparison of Damped Motion Regimes

| Parameter          | Overdamped<br>( $\gamma > \omega_0$ ) | Critically Damped<br>( $\gamma = \omega_0$ ) | Underdamped<br>( $\gamma < \omega_0$ )            |
|--------------------|---------------------------------------|----------------------------------------------|---------------------------------------------------|
| Roots of Char. Eq. | Real and distinct.                    | Real and equal<br>( $r = -\gamma$ ).         | Complex conjugate<br>( $-\gamma \pm i\omega_d$ ). |
| Motion Nature      | Non-oscillatory, slow return.         | Non-oscillatory, fastest return.             | Oscillatory with decaying amplitude.              |
| Frequency          | No frequency of oscillation.          | No frequency of oscillation.                 | $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$ .       |
| Practical Use      | Heavy door closers.                   | Galvanometers, vehicle suspension.           | Tuning circuits, musical strings.                 |